

Ecosystem modelling of the Baltic Sea: Evaluating the application of Dynamic Bayesian Networks in evolving ecosystems:

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29th August 2025

Word Count: 3677

Abstract

The silent victims of climate change, more specifically human-induced climate change, are marine ecosystems. A prime example is the Baltic Sea. With an approximate age of just 9,000 years old, the Baltic Sea has undergone drastic transformations like regime shifts because of anthropogenic stressors. Ecosystem modelling of the Baltic Sea has produced positive reversals such as the reappearance of top predators and reduction in nutrient load. However, this is not enough. By modelling ecosystems which include temporal dependencies, a greater understanding of interactions between species and its environment can be obtained. Therefore, this dissertation explores and evaluates the implementation of Dynamic Bayesian Networks (DBNs) to model the Baltic Sea sub-basins. This study is a continuation from the paper by Uusitalo et al [1] where regime shifts in the Gotland Basin were modelled using DBNs. To detect shifts between learned ecosystem states, a Hidden Markov Model (HMM) was created. DBNs are extensions of hidden Markov models which do incorporate changes in space and time, unlike the HMM. After the DBN models were produced, predicted values were computed using training and test data. To evaluate the performance of the DBN models, the root mean square error (RMSE) was calculated. Consequently, the DBN models created varying results predicting well in the southern sub-basins and not so well in the northern basin. In conclusion, DBNs were partially effective in producing accurate, predicted results. Future work can include modelling more fish species which may improve this result.

Chapter 1

Introduction

Marine ecosystems worldwide are deteriorating at an accelerating rate, particularly over the last 75 years. Unfortunately, the leading cause is anthropogenic climate change, such as increasing greenhouse gas emissions and waste pollution [2]. Marine ecosystems are vital to sustaining life on Earth. They provide natural shoreline protection, food and medicinal products, support livelihoods and most importantly, regulate the climate [2]. Yet only eight percent of the world's oceans (approximately) are classified as marine protected areas; from which, only 2.8% are sufficiently protected [3]. To effectively protect marine ecosystems, we need to understand their complex dynamics and how they interact with one another [4]. This chapter introduces the Baltic Sea, often referred to as a marine 'time machine' [5].

The Baltic Sea is one of the most severely affected marine ecosystems in the world, despite only being formed after the last glaciation. It has endured the harsh effects of acidification, deoxygenation, and warming sea levels. This has led to eutrophication, habitat loss, and species translocation [5]. It is managed by several international and multinational authorities such as the Helsinki Commission (HELCOM) where positive outcomes such as the reduction in nutrient load and partial recovery of fish stock were recorded. Therefore, the Baltic Sea can serve as a good model region for other marine areas globally which are also projected to experience similar coastal perturbations by 2100 [5].

The Baltic Sea is truly a unique marine environment due to its location and features. Unlike open oceans, the Baltic Sea is situated on a continent making it a semi-enclosed, shallow sea [6]. It is one of the largest brackish water bodies in the world. Brackish water is a mixture of saline water and freshwater. Towards the north, cold (nearly) freshwater can be found and warm, saline water in the south [7]. Due to its almost enclosed form, the Baltic Sea receives high-salinity water from the North Sea through the narrow and shallow Danish straits [8]. This limits water exchange and creates a water renewal of approximately 30 years. This leads to water stagnation that develops into hypoxic areas near the seabed as oxygen is used up by marine organisms. The Baltic Sea also has a drainage basin four times larger than its entire area causing nutrient pollution. These characteristics force the sea to have limited biodiversity, become vulnerable to pollution, and undergo regime shifts [6, 9].



Figure 1.1: Map of the Baltic Sea showing its sub-basins, drainage basin, major rivers and cities. Image from [10].

Regime shifts (RSs) are defined as abrupt, persistent transformations in an ecosystem. Regime shifts occur due to natural and anthropogenic environmental stressors such as temperature fluctuations and eutrophication [11]. Eutrophication is the enrichment of water bodies with excessive organic nutrients, particularly nitrogen and phosphorus. In the Baltic Sea, this occurred due to the improper disposal of agricultural and sewage waste [11]. Eutrophication of the Northern Baltic Sea resulted in altered nitrogen-to-phosphorus ratio, increased sedimentation rates, and increased input of organic matter to phytoplankton and algae (benthic system). An increase in biomass of the benthic system reduced water transparency and depleted oxygen reserves which had adverse effects on zooplankton and fish [12].

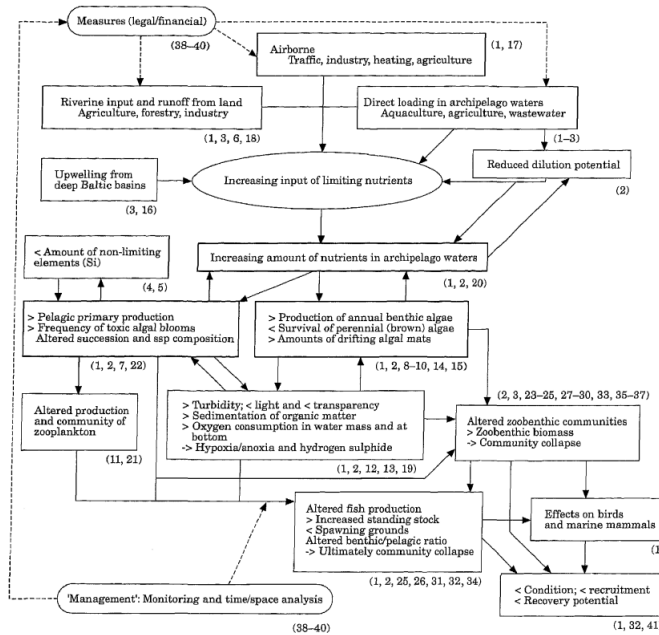


Figure 1.2: A flow model of the eutrophication process in the Baltic Sea. Diagram from [12].

Another primary driver of regime shifts in the Baltic Sea was overfishing. Fishing in the Baltic Sea was dependent on the fish species which controlled the ecosystem during that specific period [13]. Between the 1970s to late 1980s, cod dominated the Central Baltic Sea ecosystem due to increased benthic organisms and ideal reproduction conditions. This completely changed after the 1990s where sprat became the dominant species. Cods became subjected to recruitment and fishery, and its benthic food were less available due to hypoxia [13]. Regime shifts alter the natural structure and functioning of ecosystems resulting in an unstable state [11].

The instability of ecosystems poses a major challenge when implementing marine management policies and forecasting ecological changes [14]. To address this issue, Bayesian networks (BNs) can be utilised which factor in this instability or uncertainty. Bayesian networks stem from Bayes' Theorem which can be defined as calculating the conditional probability of event A given that event B has occurred [15].

$$P(A | B) = \frac{P(B | A) \cdot P(A)}{P(B)}$$

A, B = events
 $P(A|B)$ = probability of A given B is true
 $P(B|A)$ = probability of B given A is true
 $P(A), P(B)$ = the independent probabilities of A and B

Figure 1.3: An image showing Bayes' Theorem from [16].

Bayesian networks are represented as graphs consisting of nodes and edges. The edges quantify the relationship between two nodes or variables through conditional probability [15]. Bayesian networks introduce hidden variables i.e. latent factors that influence ecosystems [1]. Therefore, BNs can be applied to macro systems such as ecosystems where there are multiple, interacting factors and natural uncertainty allowing more accurate inferences to be made.

A limitation of this model is that it does not account for dynamic relationships taking place in various temporal and spatial scales [17]. Dynamic Bayesian Networks (DBNs) are extensions of Bayesian networks which incorporate BNs when modelling time series data. Time slices are used to show the system at different points in time representing dynamic processes [18]. Dynamic Bayesian Networks can therefore provide a more practical marine management framework and make predictions from this.

Uusitalo et al [1] successfully demonstrates DBNs to model regime shifts in the Gotland Basin, Baltic Sea. Thus, the primary aim of this dissertation is to evaluate the broader applicability of DBNs in other Baltic Sea sub-basins focussing on fish recruits. This study will also identify the main drivers of ecosystem dynamics in the Baltic Sea and how these ecosystem dynamics change over time and within different

sub-basins.

This literature review synthesises key findings across multiple disciplines offering a comprehensive overview of why the Baltic Sea needs to be modelled and the prospective role of DBNs.

Chapter 2

Data and Methods

2.1 Data

The data provided were environmental and biological monitoring data which represented food-web dynamics in five Baltic Sea sub-basins from the years 1966 – 2023. The data were acquired from ‘a wide-range of open access sources’ collected by the Finnish Environmental Institute (SYKE), the Swedish Meteorological and Hydrological Institute (SMHI), International Council for Exploration of the Sea (ICES), the Leibniz Institute for Baltic Sea Research (IOWDB), and HELCOM [19]. The biotic variables in this dataset were chosen to represent key, prevailing ecosystem components from each sub-basin.

Fish recruitment refers to the process where young fish survive to adulthood and become vulnerable to harvesting [20]. In the dataset, there were several types of fish data available such as recruitment, biomass, and spawning biomass. Fish recruitment was chosen because this was the only variable consistent in each sub-basin dataset.

Bothnian Bay
Bothnian Sea
Gulf of Finland
Central Baltic
Western Baltic

Table 2.1: The 5 Baltic Sea sub-basins.

Herring
Cod
Sprat
Vendace
Plaice

Table 2.2: Fish species in the Baltic Sea.

Abiotic Variables	Definition
SAL_win	Winter surface salinity
pH	
DIP	Dissolved Inorganic Phosphorus
DIN	Dissolved Inorganic Nitrogen
TP	Total Phosphorus
Si	Silicate
NH4	Ammonium
NO2	Nitrogen Dioxide
NO2	Nitrate
DOC	Dissolved Organic Carbon
SHT_su	Hypolimnion temp (June - August)
SST_su	Sea surface temperature (June - August)
SST_spr	Sea surface temperature (April - May)
TN	Total Nitrogen
SAL_deep	Winter bottom salinity
DIP_deep	Winter bottom Dissolved Inorganic Phosphorus
TP_deep	Winter bottom Total Phosphorus
TN_deep	Winter bottom Total Nitrogen
Ox_deep	Winter bottom oxygen
DIN_deep	Winter bottom Dissolved Inorganic Nitrogen
SpringBloom	Total phytoplankton biomass
Ice_max	Whole Baltic extent of ice
N.load	Nitrogen loads + atmospheric deposition
P.load	Phosphorus loads
Anoxic	Anoxic area < 0 ml/lO ₂
Hypoxic	Anoxic area < 2 ml/lO ₂
WBSI	Winter Baltic Sea Index

Table 2.3: Abiotic Variables and its definitions

2.2 Methods

The methods described below were used in the analysis of all sub-basins.

2.2.1 Explanatory Data Analysis (EDA)

To prepare the data for analysis, all variables were log-transformed using ‘log(.+1)’ to avoid log of zero. Missing data values had to be explicitly changed into ‘NA’

values in R. The ‘NA’ values were retained to preserve the integrity of the data. Correlation analysis was performed using Spearman’s rho method as the raw data was non-linear. Scatter plots showing each fish recruit against each abiotic variable were produced using the R package “ggplot2” to explore initial patterns. A correlation heatmap was created to visualise the strength and direction of these bivariate variables [21].

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2-1)}$$

Figure 2.1: Spearman’s rho formula, where ρ = Spearman Correlation Coefficient, rank = position of the variable’s value relative to the other values in the dataset, ρ_i = difference in ranks between two variables of each data value and n = total number of observations [21].

2.2.2 Hidden Markov Model (HMM)

The ‘NA’ values for the main analysis were imputed using Predicted Mean Matching (PMM) method from the “mice” package. The PMM method involves predicting values of the target variable, including the missing values. For each missing value, the closest predicted value to the predicted value of the missing value was selected. Finally, the actual value of that predicted value is used to replace the missing value [22]. The data for this sub-section was standardised to a mean of zero and standard deviation of one to achieve normality and allow parameter learning [1].

To identify the underlying regime shifts, a Hidden Markov Model was created using the package “depmixS4”. The Markov Model is a stochastic process which states that the future state of an event is solely dependent on its present state, not its past states [23]. The HMM is formed based on the Markov Model. It consists of two sequences: observation sequences and an underlying sequence which is estimated from the observed sequences [24].

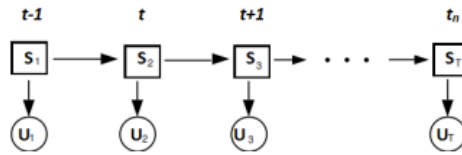


Figure 2.2: A Hidden Markov Model showing the square nodes as hidden variables and circle nodes as observed variables. Diagram from [25].

A 2-state multivariable HMM was made using the relevant fish recruit and abiotic variables. The hidden variable can only take two values and is arbitrary; the shifts between state 1 and 2 (and vice versa) is desired. A fixed random seed was set to the model for reproducibility. The model was fitted using the Viterbi algorithm to make parameter estimations. This finds the highest probable sequence of hidden states [26]. The hidden states were plotted in a line graph to learn the potential regime shifts. A limitation of the model is the autoregressive (AR) link. This means

that the first prediction (and consequently the rest of the predictions) calculated would be less accurate due to the lack of previous observations. It regresses on itself [25].

Parameter Estimates	Definition
Initial State probabilities	Probability distribution of the initial hidden state
Transition probabilities	The probability of moving from one hidden state to another
Emission probabilities	The probability of an observation at a hidden state

Table 2.4: Parameter estimations provided after the HMM is fitted and its definitions [27].

2.2.3 Dynamic Bayesian Network (DBN)

Continuing from the Introduction, DBNs are presented in a directed acyclic graph (DAG) – a directed graph with no cycles. Two time intervals were shown; t were the lagged variables and $t + 1$ were the “current” variables/observations. Each edge starts from a parent node to a child node [18].

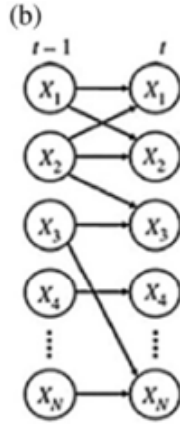


Figure 2.3: A Dynamic Bayesian Network showing circle nodes as variables at two different time slices. Image from [28].

The DBN model was constructed using the “bnlearn” package. A DBN was created for each sub-basin and each fish recruit. It modelled the hidden state, a particular fish recruit and its abiotic variables at two different, time intervals. The two highest, positive correlation coefficients were selected from the heatmaps. This was in alignment with the previous work from Uusitalo et al [1]. An exception was the DBN model created with cod recruits where the relationships were chosen based on the paper by Uusitalo et al [1]. The structured model was then fitted and a DBN plot was created using the “Rgraphviz” package.

Predicted values were generated from the DBN model by splitting 70% of the data into training data and the remaining 30% as test data. Graphs showing the actual against predicted data were presented. To evaluate the performance of the model, the root mean square error (RMSE) was computed. The RMSE is measure by calculating the difference between the actual and predicted values, residuals. The residuals are squared and summed together. This value is divided by the degrees of freedom of the model, $N - P$, calculating the mean squared residual or mean squared error (MSE). Finally, the square root of MSE is taken. It measures how dispersed the residuals are from the regression line, quantifying the average size of the prediction error [29].

The RMSE values were rounded to four significant figures. The formula for RMSE:

$$RSME = \sqrt{\frac{\sum (y_i - \hat{y}_i)^2}{N - P}}$$

Figure 2.4: An image showing the RMSE formula where y_i is the actual value of the i th observation, $\hat{y}_i$ is the predicted value of i th observation, n is the number of observations and p is the number of parameter estimates [29].

Chapter 3

Main Results

3.1 Explanatory Data Analysis (EDA)

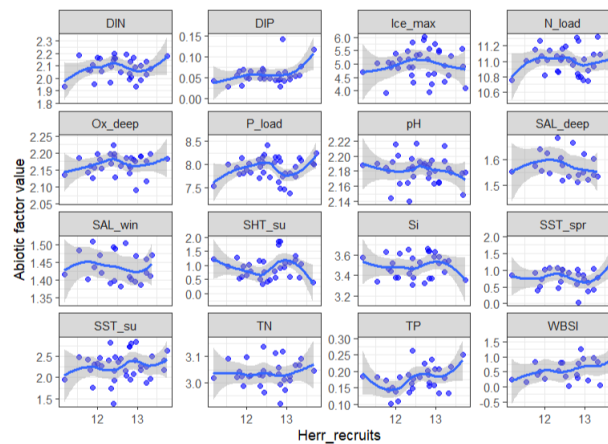


Figure 3.1: Scatter plots showing each abiotic variables against herring recruits in the Bothnian Bay.

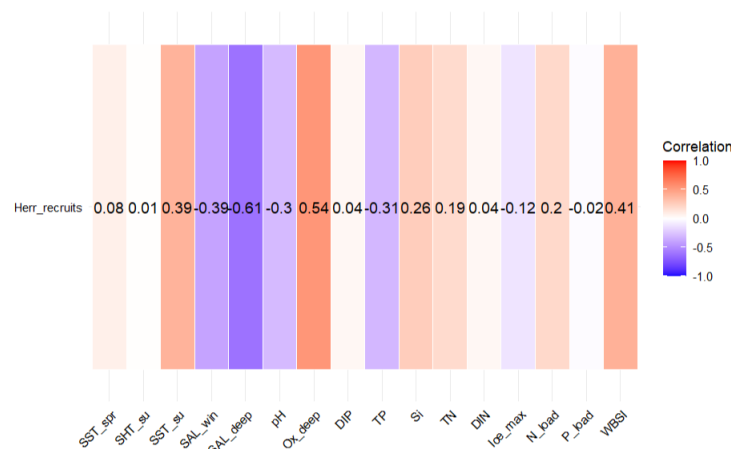


Figure 3.2: A correlation heatmap showing the respective correlation coefficients calculated between herring recruits and abiotic variables for the Bothnian Bay.

The scatter plots show the raw data as blue points, the blue curve as the line of best fit and the grey area as the confidence interval of the fitted curve. In figure 3.1, over half of the abiotic variables have a weak positive relationship while others have a weak negative or no, clear relationship with herring recruits. Most of the data points are randomly scattered away from the curve except for the abiotic variable, DIP, where nearly all the data points are clustered around the curve. The correlation heatmap, figure 3.2, describes the strength and direction of relationships using a colour gradient. Blue (1) indicates the strongest, positive relationship and red (-1) indicates the strongest, negative relationship. Most of the abiotic variables are positively correlated to herring recruits. The two strongest, positive coefficients are from Ox_deep and WBSI.

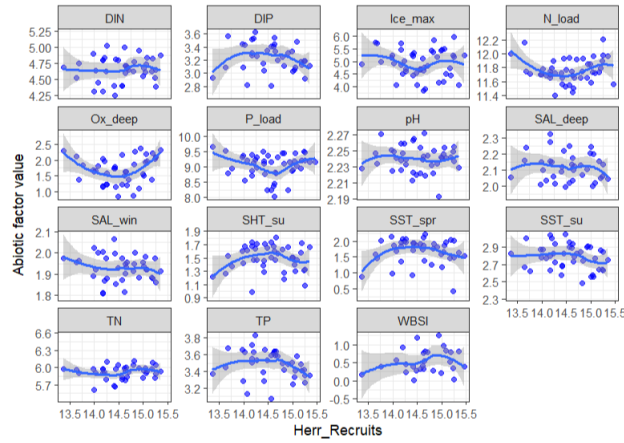


Figure 3.3: Scatter plots showing each abiotic variables against herring recruits in the Gulf of Finland (GoF).

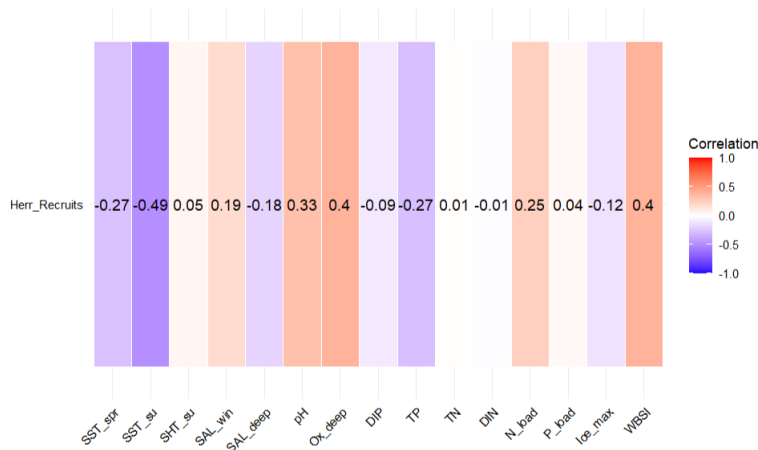


Figure 3.4: A correlation heatmap showing the respective correlation coefficients calculated between herring recruits and abiotic variables for the Gulf of Finland.

The scatter plots in figure 3.3 demonstrate that nearly half of the abiotic variables share a weak negative relationship with herring recruits in the GoF. The plot showing TN has data points close to the curve; the other plots do not. The results in figure 3.4 reveal only three abiotic variables being moderately, positively correlated to herring recruits. The two strongest, positive coefficients are from Ox_deep and WBSI.

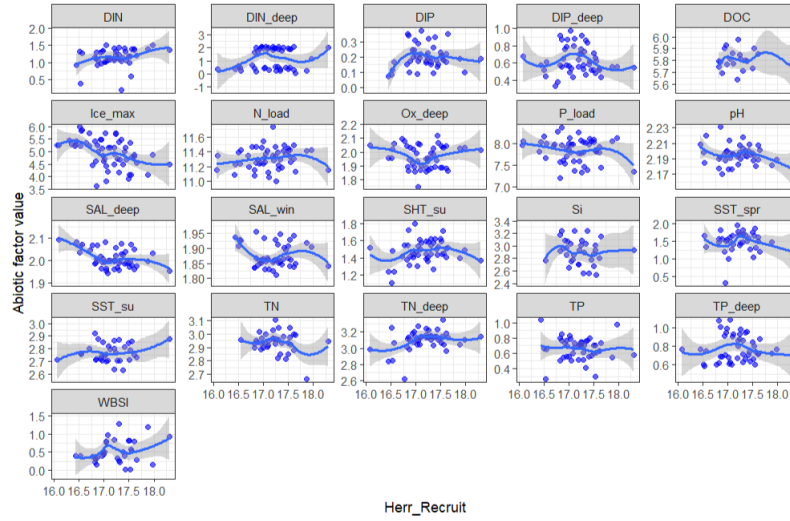


Figure 3.5: Scatter plots showing each abiotic variables against herring recruits in the Bothnian Sea.

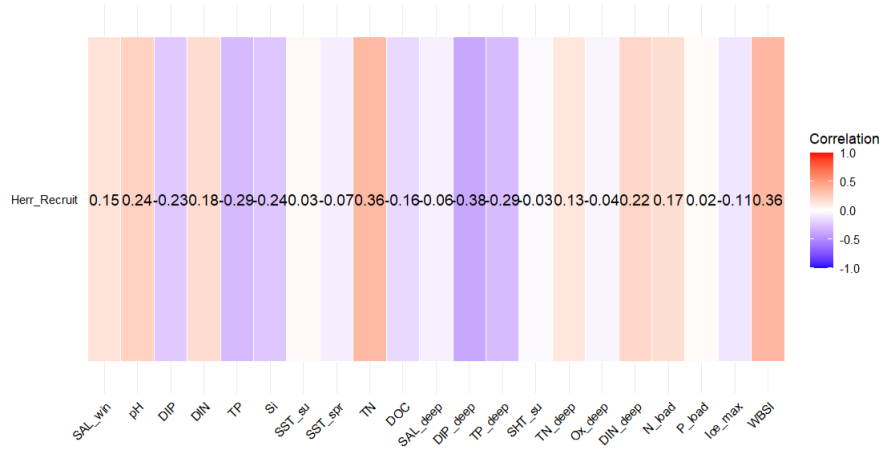


Figure 3.6: A correlation heatmap showing the respective correlation coefficients calculated between herring recruits and abiotic variables for the Bothnian Sea.

Figure 3.5 shows that ten plots have a weak negative relationship with herring recruits in the Bothnian Sea. Majority of the data points in the plots are randomly distributed towards the centre of the curve. Only three abiotic variables demonstrate no correlation with herring recruits in figure 3.6. The two strongest, positive coefficients are TN and WBSI.

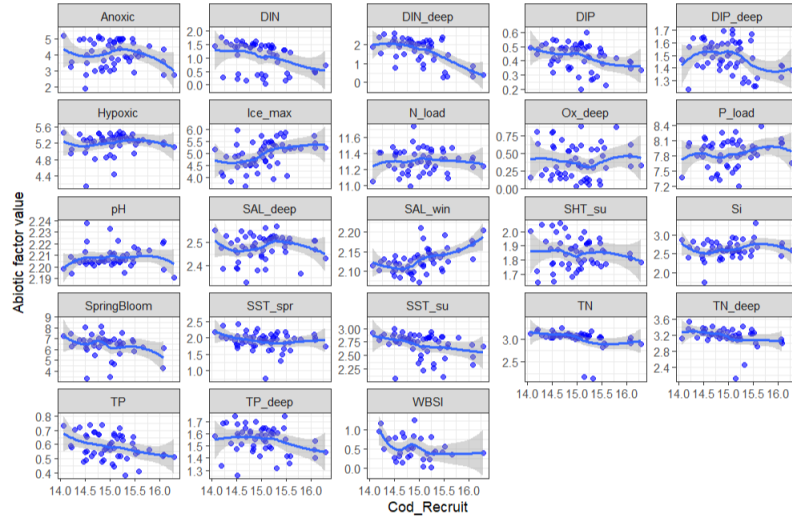


Figure 3.7: Scatter plots showing each abiotic variables against cod recruits in the Central Baltic.

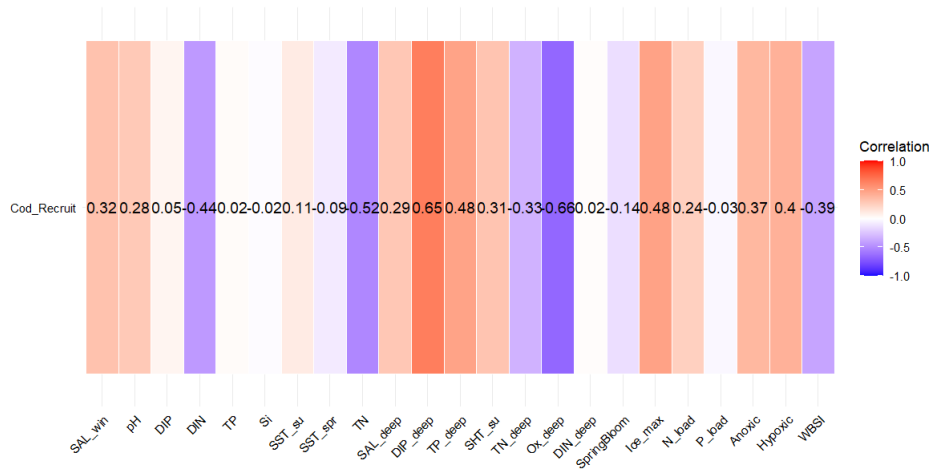


Figure 3.8: A correlation heatmap showing the respective correlation coefficients calculated between cod recruits and abiotic variables for the Central Baltic.

The scatter plots in figure 3.7 show a mix of weak negative and weak positive relationships between cod recruits and abiotic variables in the Central Baltic. It appears as if the data points are randomly scattered towards the left side to centre of each plot. Figure 3.8 reveals over half of the abiotic variables having a positive or no correlation with cod recruits. The only fish recruit demonstrating the most positive correlation coefficients out of all the sub-basins and fish species.

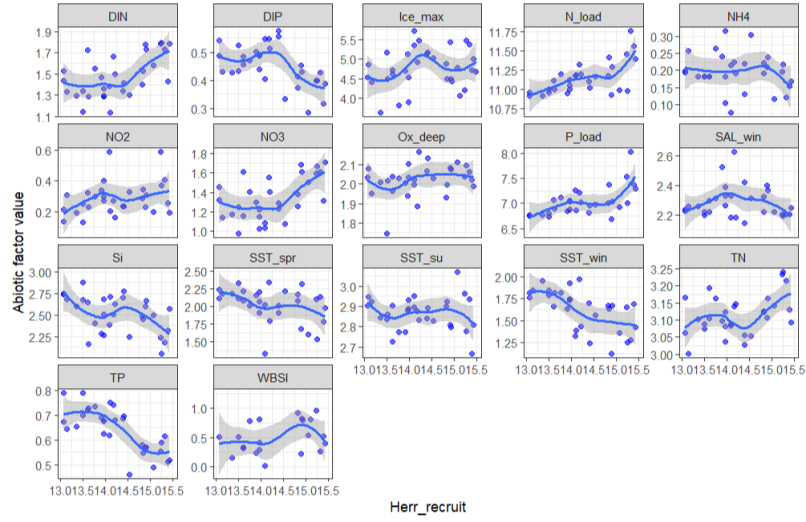


Figure 3.9: Scatter plots showing each abiotic variables against herring recruits in the Western Baltic.

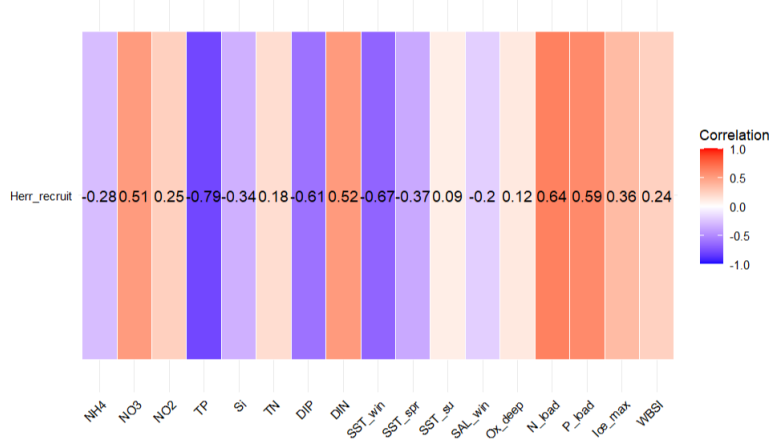


Figure 3.10: A correlation heatmap showing the respective correlation coefficients calculated between herring recruits and abiotic variables for the Western Baltic.

Figure 3.9 shows an almost equal number of abiotic variables showing a weak negative or weak positive relationship. The data points are randomly distributed throughout the plots. Like figure 3.9, the correlation heatmap (figure 3.10) shows a nearly equal amount of positive and negative correlations with zero abiotic variables showing no correlation. The two strongest, positive coefficients are N_load and P_load.

3.2 Hidden Markov Models (HMMs)

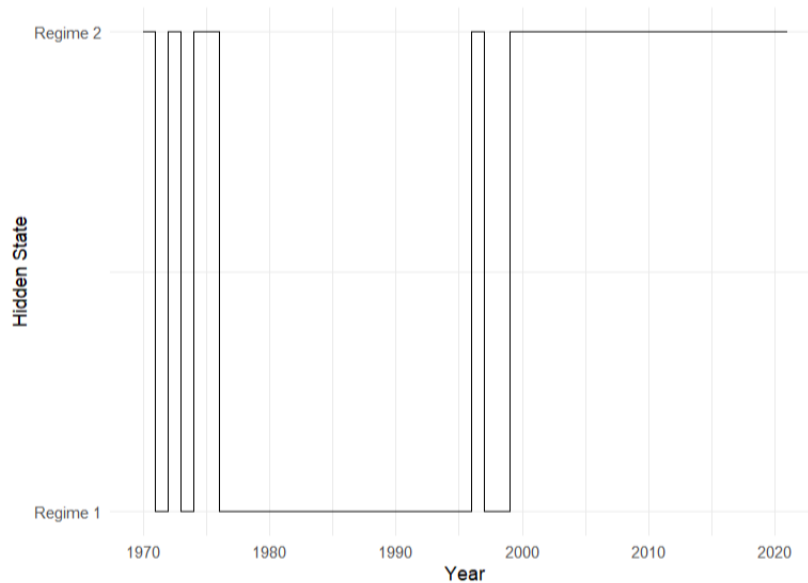


Figure 3.11: Regime shifts discovered in the Bothnian Bay using a HMM.

Figure 3.11 shows the learned ecosystem states between 1970 to 2021 in the Bothnian Bay. Shifts between ecosystem states are observed between the early 1970s to mid-1970s. This may correspond with the work by [13].

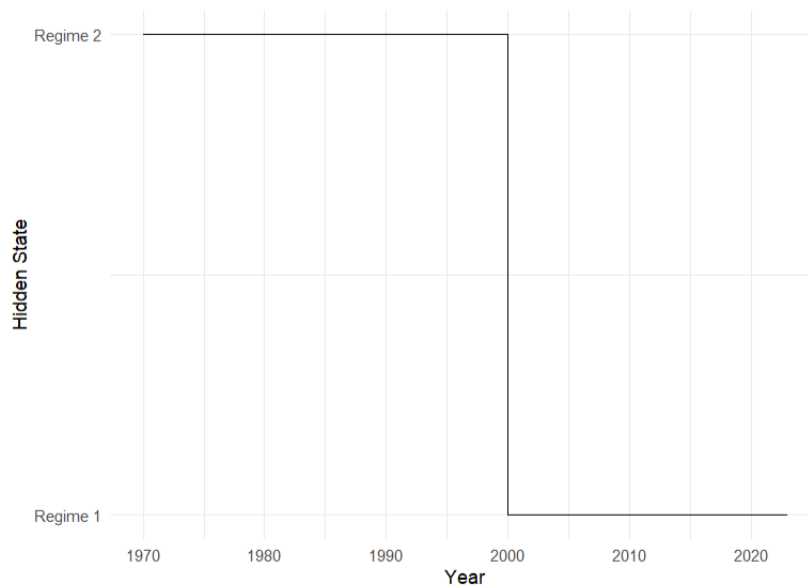


Figure 3.12: Regime shifts discovered in the Gulf of Finland (GoF) using a HMM.

Figure 3.12 displays learned ecosystem states identified in the GoF between 1970 to 2023. A regime shift is detected in the year 2000.

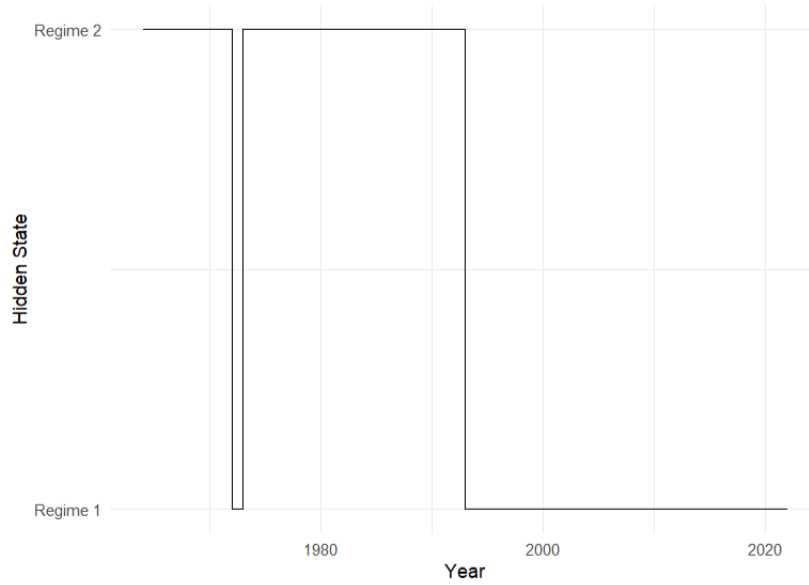


Figure 3.13: Regime shifts discovered in the Bothnian Sea using a HMM.

Figure 3.13 shows the learned ecosystem states between 1964 to 2022 in the Bothnian Sea. A shift in states is noted, around 1976 and 1977. Another shift is noted in the 1990s. This aligns with the regime shift written in the paper by [1].

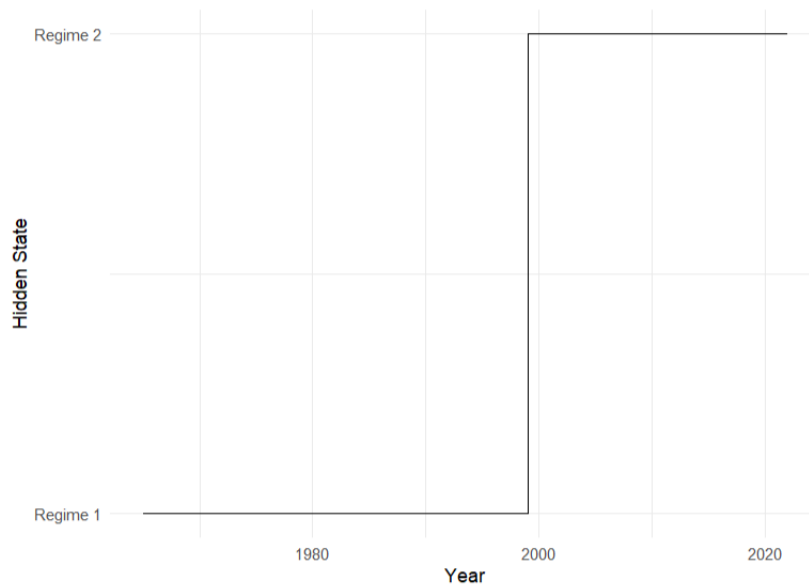


Figure 3.14: Regime shifts discovered in the Central Baltic using a HMM.

Figure 3.14 displays the learned ecosystem states between 1965 to 2022 in the Central Baltic. A regime shift is detected in the late 1990s.

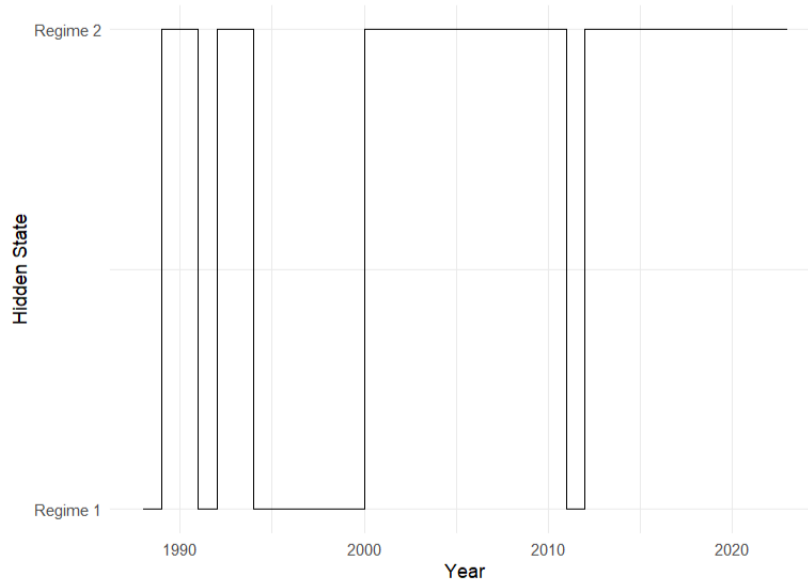


Figure 3.15: Regime shifts discovered in the Western Baltic using a HMM.

Figure 3.15 displays the learned ecosystem states between 1988 to 2023 in the Western Baltic. Quite a few shifts in ecosystem states were noted in this time period. Notable regime shifts can be observed in the late 1980s to the early 1990s. This coincides with the regime shifts mentioned in the papers by [13] and [1].

3.3 Dynamic Bayesian Network Models (DB-NMs) and predictions

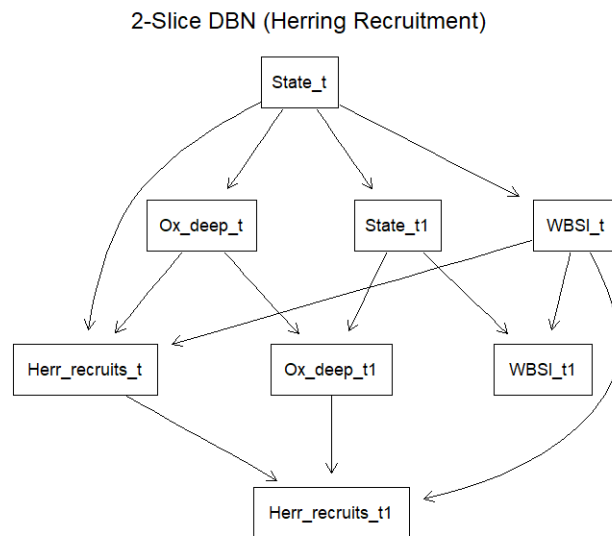


Figure 3.16: A prescribed DBN model of the Bothnian Bay.

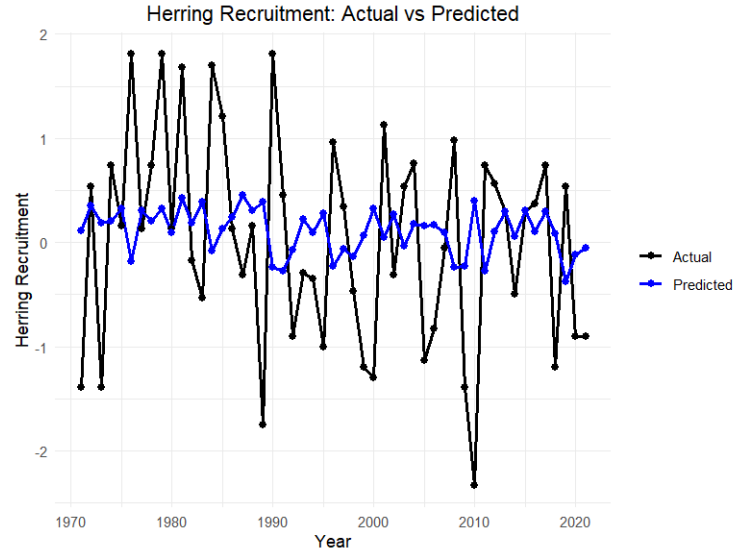


Figure 3.17: A line graph displaying the actual vs predicted values for herring recruits calculated by its respective DBN model.

The DBN model in figure 3.16 shows the relationships between hidden states, herring recruits, Ox_deep and WBSI across time slices in the Bothnian Bay.

In figure 3.17, the predicted values are shown close together around zero of the y-axis. However, the actual values are varied with a range of four, approximately. Some of the points are in the opposite range to the actual values e.g. the data points between 1975 to 1980 and in 2010. The RMSE for this model is equal to 1.066 (4 sf).

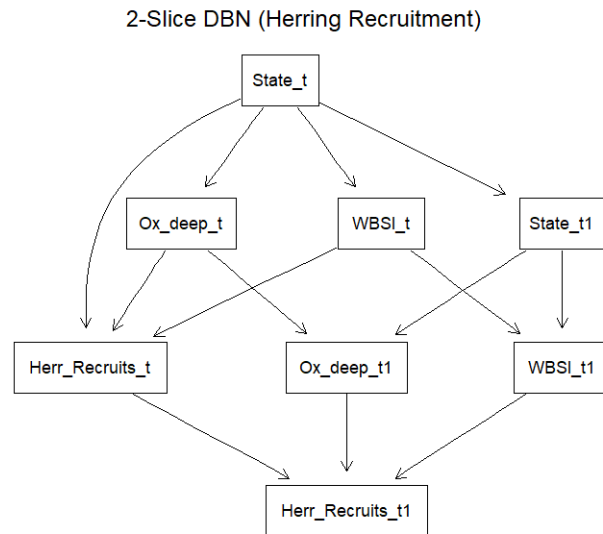


Figure 3.18: A prescribed DBN model of the Gulf of Finland (GoF).

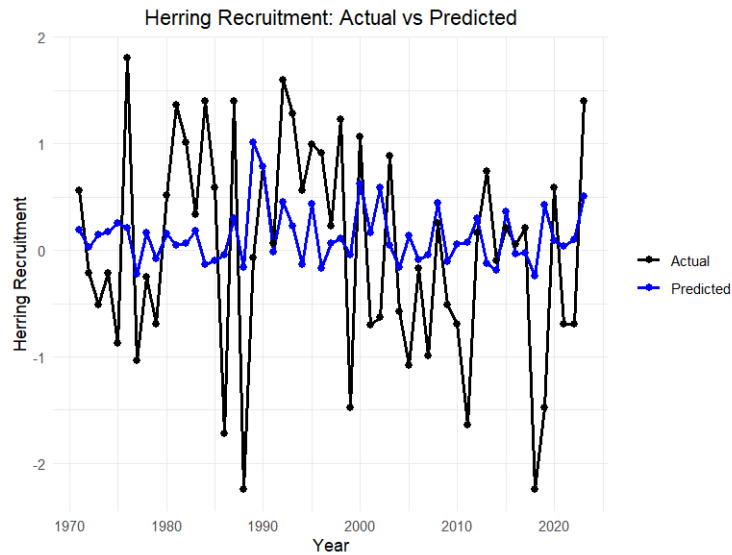


Figure 3.19: A line graph displaying the actual vs predicted values for herring recruits calculated by its respective DBN model.

The DBN model in figure 3.18 shows the relationships between hidden states, herring recruits, Ox_deep and WBSI across time slices in the GoF. Similar to the sub-basin above, the predicted values in figure 3.19 hover around the middle of the graph. The actual values are more dispersed. The model does predict some of the values well e.g. around the 1990s, the same shape is seen in the actual values only predicted a few years earlier. However, the RMSE value for this basin is 0.9593 (4 sf).

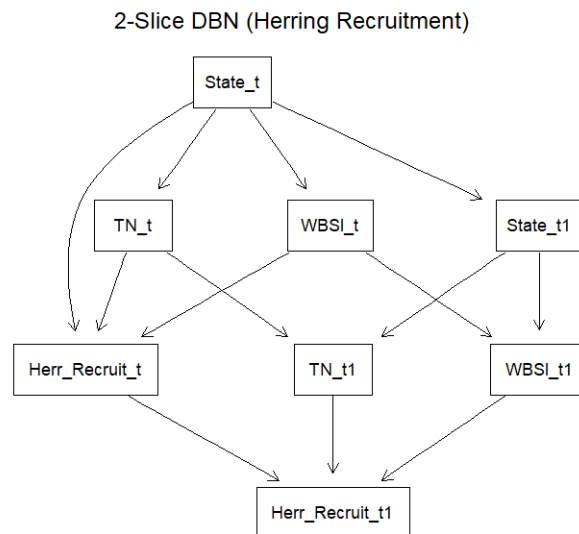


Figure 3.20: A prescribed DBN model of the Bothnian Sea.

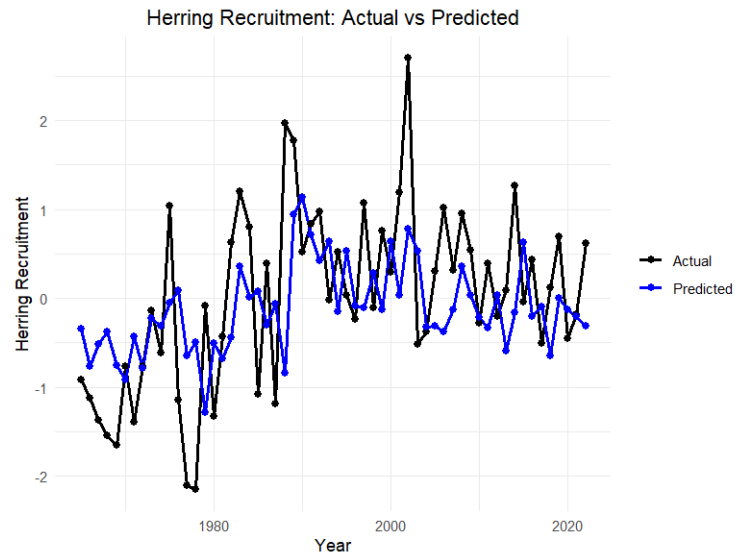


Figure 3.21: A line graph displaying the actual vs predicted values for herring recruits calculated by its respective DBN model.

The DBN model in figure 3.20 shows the relationships between hidden states, herring recruits, TN and WBSI across time slices in the Bothnian Sea.

The predicted values in figure 3.21 seem to marginally better than the previous two sub-basins. The direction of the predicted values was reproduced to a certain extent, but the magnitude is significantly smaller than the actual values. The calculated RMSE value is 0.9108 (4 sf).

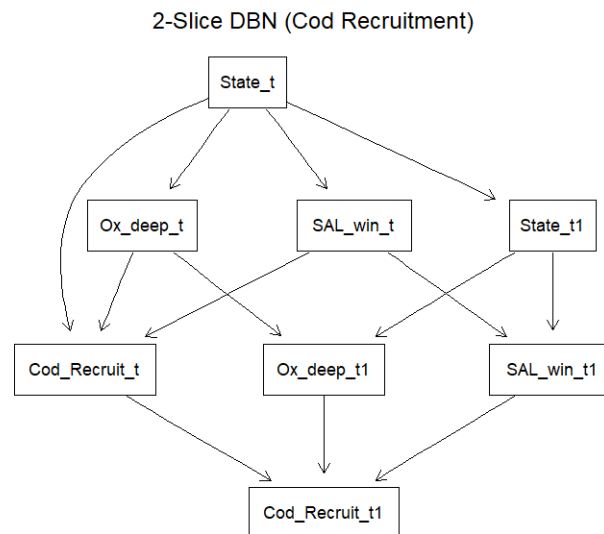


Figure 3.22: A prescribed DBN model of the Central Baltic.

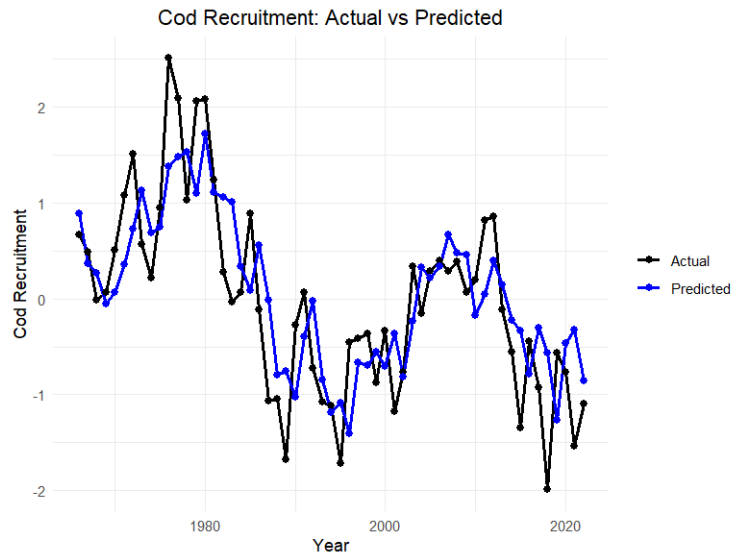


Figure 3.23: A line graph displaying the actual vs predicted values for cod recruits calculated by its respective DBN model.

The DBN model in figure 3.22 shows the relationships between hidden states, cod recruits, Ox_deep and Sal_win across time slices in the Central Baltic. The predicted values in figure 3.23 are vastly better than previous, predicted values in terms of direction and size. However, a difference can be observed in the magnitude where the prediction is significantly smaller. The autoregressive lag is evident in this graph. The RMSE for this model is 0.6113 (4 sf).

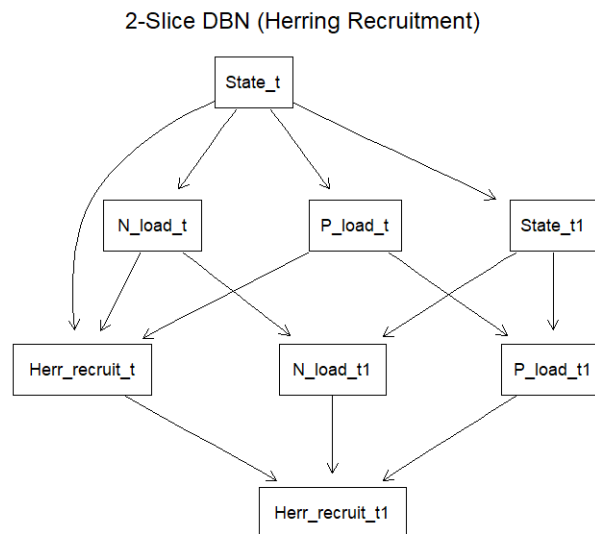


Figure 3.24: A prescribed DBN model of the Western Baltic.

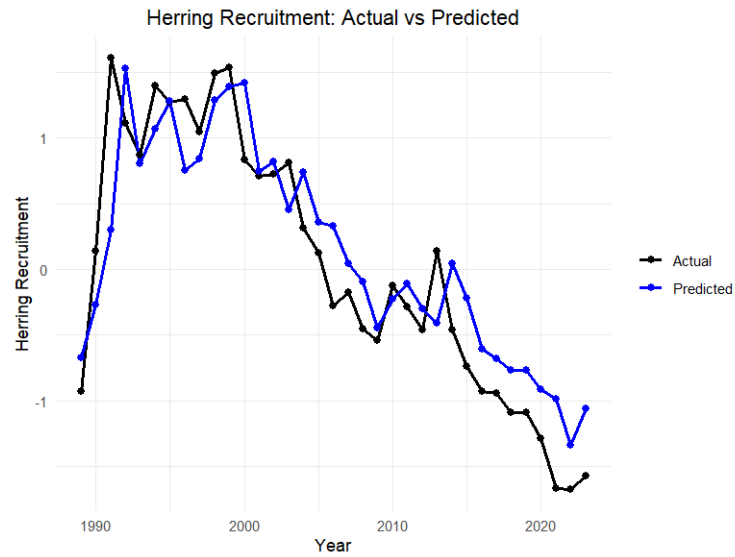


Figure 3.25: A line graph displaying the actual vs predicted values for herring recruits calculated by its respective DBN model.

The DBN model in figure 3.24 shows the relationships between hidden states, herring recruits, N_load and P_load across time slices in the Western Baltic.

The graph in figure 3.25 shows the best fitted model in the whole study. The direction of the actual values was captured well. But there are significant differences in magnitude, although smaller than the previous models. The autoregressive lag is evident in this graph. The RMSE value for this sub-basin is equal to 0.4190 (4 sf).

Chapter 4

Discussion

The primary objective of this report was to evaluate the application of DBNs in other Baltic Sea sub-basins. This was evaluated by calculating the RMSE value of each model. The key finding from the models' performances was that it was highly varied and dependent on the location and abiotic variables of each sub-basin.

4.1 DBN models

The Bothnian Bay produced the poorest evaluation results with a RMSE value of 1.066 (4 sf). This suggests a poor model fit. The Bothnian Bay is the northernmost basin in the Baltic Sea indicating that it mainly consists of well-oxygenated, freshwater. The water turnover in this area is around seven years. It has a well-mixed water column with suitable oxygen conditions and is not affected by nutrient loading. This implies that it is less likely to be affected by eutrophication in comparison to the other basins [30].

The next sub-basin, the Gulf of Finland, generated a RMSE value of 0.9593 (4 sf). This is a high value suggesting a poorly fitted model. This basin is most affected by eutrophication out of all Baltic Sea sub-basins. It has poor oxygen conditions and has eight times more nutrient load than the Gulf of Bothnia [30].

The RMSE value for the Bothnian Sea was 0.9108 (4 sf). This is also quite a high value suggesting a poor model fit. Although, the Bothnian Bay and Bothnian Sea are in the same gulf; eutrophication affects the Bothnian Sea more because of its location. It is nearer the centre of Baltic Sea. Consequently, it has a salinity around four psu (practical salinity unit) higher than the Bothnian Bay. Its inner areas have a mean TP value of $26 \mu\text{g L}^{-1}$ [30].

The Central Baltic produced an improved RMSE result of 0.6113 (4 sf). However, this indicates a poor-to-moderately fitted model. Tomczak et al [13] described that cod numbers collapsed after the 1990s due to insufficient food availability and overfishing. During this period, there was an increase in eutrophication which did not support the benthic organisms' livelihood.

The model's optimal performance was from the Western Baltic. It produced the lowest RMSE value of 0.4190 (4 sf) indicating a moderately fitted model. Kuss et

al [31] reported that the amount of anthropogenic nitrogen in the Western Baltic Sea decreased by about 30% since 1995. However, it also reported that this did not bring about any significant drops in eutrophication levels.

4.2 Model error and limitations

Whilst the model was partially successful in predicting values for the Central and Western Baltic sub-basin, there was a potential error. Figure 11 shows shifts in the learned ecosystem states in the Bothnian Bay. Around four shifts were observed in a span of five years (1970 to 1975). In theory, this could be the actual case, but it is more likely to have occurred due to noise.

A limitation of this study was that there were a significant number of missing values in each sub-basin dataset which were imputed using PMM. This affects the validity of the model. In figures 24 and 25, a visible lag can be seen in the predicted values. This was caused by the autoregressive link which affects the accuracy of the model. Another limitation of this model was the inconsistency of fish variables in all sub-basin datasets. Consequently, variables such as fish biomass could have been used as a better representative of fish populations. Additionally, this also caused only two fish species, herring and cod, to be modelled. If more fish species were used in this model, a more accurate representation of the ecosystem would have been produced.

4.3 Conclusions and future work

Overall, utilising DBNs to model ecosystems is an emerging method. However, it may be reliant on certain ecosystem traits such as eutrophication. As seen in the results section, more accurate models were produced towards the southern regions where eutrophication increases. Data is predicted more precisely in unstable, eutrophic states compared to slightly more stable, less eutrophic states. The Gulf of Finland is an exception because it is mostly surrounded by land resulting in most of the excess nutrients to seep into the basin [32]. From the EDA, it can be deduced that the main drivers of ecosystem dynamics in the Baltic Sea are anthropogenic stressors such as excess waste and overfishing. Each of the Baltic sub-basins have experienced a shift in ecosystem states because of anthropogenic climate change over the past 65 years, with southern basins being affected the most.

Future research could consider using the Autoregressive Hidden Markov Model instead of the HMM to avoid the time lag. DBN models studying key (fish) stock variables and a broader range of fish species could also be applied. Furthermore, extending DBN models to examine key species at each trophic level of the Baltic Sea could enhance our understanding of its ecosystem dynamics.

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